

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design and Implementation of a Styled Webpage Using CSS3 Advanced Styling and HTML5 Canvas 2D Graphics

2. AIM

To design and develop an interactive modern webpage utilizing advanced CSS3 styling features (linear gradients, flexbox multi-column layout, rounded borders, box shadows, text shadows, hover transitions) and HTML5 Canvas JavaScript 2D rendering API to draw geometric primitives (rectangles, circles, lines) and custom typography.

3. OBJECTIVE

The objectives of this experiment are:

- To implement modern CSS3 background gradient effects (linear-gradient).
 - To build a responsive flexbox multi-column card layout using display: flex, gap, and flex: 1.
 - To apply aesthetic CSS3 properties including box-shadow, border-radius, and text-shadow.
 - To implement interactive pseudo-classes (:hover) for dynamic button and card transitions.
 - To understand the HTML5 <canvas> element and obtain its 2D rendering context (getContext('2d')).
 - To programmatically render 2D graphics using JavaScript methods: fillRect(), arc(), moveTo(), lineTo(), stroke(), and fillText().
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4. THEORY

1. CSS3 Styling & Layout:

- **Linear Gradients:** linear-gradient(direction, color1, color2) generates smooth multi-color transitions without image files.
- **Flexbox Layout:** display: flex creates a flexible container. gap defines spacing between items, and flex: 1 distributes equal widths to children.
- **Visual Styling:** border-radius softens corners, box-shadow gives elevation/depth, and text-shadow adds emphasis to headings.
- **Hover Effects:** :hover triggers dynamic styling modifications when the mouse hovers over interactive elements.

2. HTML5 Canvas 2D API:

- The <canvas> element defines a bitmapped drawing surface in HTML.
- canvas.getContext("2d") provides drawing methods and properties.
- **Rectangle:** ctx.fillRect(x, y, width, height) renders solid colored boxes.
- **Circle / Arc:** ctx.arc(x, y, radius, startAngle, endAngle) creates circular paths.
- **Line Path:** ctx.moveTo(x1, y1) and ctx.lineTo(x2, y2) define paths rendered via ctx.stroke().
- **Typography:** ctx.font and ctx.fillText("text", x, y) draw custom text directly on the canvas.

CSS3 Background & Container → Header & Nav → HTML5 Canvas 2D Rendering → Flexbox Cards → Hover Transitions → Footer

5. ALGORITHM / PROCEDURE

1. Create a new directory and create an index.html file.
 2. Set up the HTML5 document structure with <head>, <style>, and <body>.
 3. Apply global styles with a linear gradient background (#d9eaff to #ffffff) and define a centralized .container with rounded corners and box shadows.
 4. Design a <header> with primary blue background (#4a6cf7) and text shadow.
 5. Create a dark navigation bar (<nav>) with interactive links having an orange hover background.
 6. Define an HTML5 <canvas id="myCanvas" width="700" height="250"> with a border and border-radius.
 7. Construct a .cards flexbox container containing three cards (HTML5, CSS3, Canvas) with action buttons and hover color inversions.
 8. Add a dark <footer> with author information: **"Created by B. Charishma"**.
 9. Write JavaScript inside <script> to access the 2D context:
 - Draw a blue solid rectangle at (50, 50) with width 150 and height 80.
 - Draw a red filled circle at center (300, 90) with radius 50.
 - Draw a green line from (430, 50) to (600, 150) with line width 5.
 - Render custom purple typography text **"B. Charishma"** at (250, 220) using 30px Arial font.
 10. Launch the page in a web browser and verify all styling and graphics rendering.
-

6. PROGRAM

index.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>My Styled Webpage</title>
```

```
<style>
```

```
body {  
    font-family: Arial;  
    background: linear-gradient(to right, #d9eaff, #ffffff);  
    margin: 20px;  
}
```

```
.container {  
    background: white;  
    padding: 20px;  
    border-radius: 10px;  
    box-shadow: 0px 5px 10px gray;  
}
```

```
header {  
    background-color: #4a6cf7;  
    color: white;  
    text-align: center;  
    padding: 20px;  
    border-radius: 10px;  
    text-shadow: 2px 2px 4px black;  
}
```

```
nav {  
    background-color: #333;  
    padding: 10px;  
    margin: 15px 0;  
    border-radius: 8px;  
}
```

```
nav a {  
    color: white;  
    text-decoration: none;  
    padding: 10px;  
    margin: 10px;
```

```
}
```

```
nav a:hover {  
    background-color: orange;  
    border-radius: 5px;  
}
```

```
canvas {  
    display: block;  
    margin: auto;  
    border: 3px solid #4a6cf7;  
    border-radius: 8px;  
}
```

```
.cards {  
    display: flex;  
    gap: 15px;  
    margin-top: 20px;  
}
```

```
.card {  
    flex: 1;  
    background-color: #eeeeee;  
    padding: 15px;  
    border: 2px solid #4a6cf7;  
    border-radius: 10px;  
    box-shadow: 0px 3px 8px gray;  
    text-align: center;  
}
```

```
.card:hover {  
    background-color: #4a6cf7;  
    color: white;  
}
```

```
button {  
    background-color: #4a6cf7;  
    color: white;  
    padding: 8px 20px;  
    border: none;  
    border-radius: 5px;  
}
```

```
button:hover {  
    background-color: green;  
}
```

```
footer {  
    margin-top: 20px;  
    background-color: #333;  
    color: white;  
    padding: 15px;  
    text-align: center;  
    border-radius: 8px;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<header>
```

```
    <h1>My Styled Webpage</h1>
```

```
    <p>HTML5 Canvas and CSS3 Styling</p>
```

```
</header>
```

```
<nav>
```

```
<a href="#">Home</a>

<a href="#">Canvas</a>

<a href="#">About</a>

<a href="#">Contact</a>

</nav>


<canvas id="myCanvas" width="700" height="250"></canvas>


<div class="cards">


  <div class="card">

    <h3>HTML5</h3>

    <p>Used to create webpage structure.</p>

    <button>Learn</button>

  </div>


  <div class="card">

    <h3>CSS3</h3>

    <p>Used for webpage styling.</p>

    <button>Learn</button>

  </div>


  <div class="card">

    <h3>Canvas</h3>

    <p>Used for drawing graphics.</p>

    <button>Learn</button>

  </div>


</div>


<footer>

  <p>Created by B. Charishma</p>

</footer>
```

</div>

<script>

```
let canvas = document.getElementById("myCanvas");
```

```
let ctx = canvas.getContext("2d");
```

```
// Rectangle
```

```
ctx.fillStyle = "blue";
```

```
ctx.fillRect(50,50,150,80);
```

```
// Circle
```

```
ctx.beginPath();
```

```
ctx.arc(300,90,50,0,Math.PI*2);
```

```
ctx.fillStyle = "red";
```

```
ctx.fill();
```

```
// Line
```

```
ctx.beginPath();
```

```
ctx.moveTo(430,50);
```

```
ctx.lineTo(600,150);
```

```
ctx.strokeStyle = "green";
```

```
ctx.lineWidth = 5;
```

```
ctx.stroke();
```

```
// Name
```

```
ctx.font = "30px Arial";
```

```
ctx.fillStyle = "purple";
```

```
ctx.fillText("B. Charishma",250,220);
```

</script>

</body>

</html>

7. CODE EXPLANATION

Property / Method	Explanation
linear-gradient(...)	Generates a smooth color gradient from light blue (#d9eaff) to white (ffffff) horizontally across the page.
box-shadow: 0px 5px 10px gray	Adds soft shadow elevation underneath container and card components.
text-shadow: 2px 2px 4px black	Applies a drop-shadow effect to the header heading for higher contrast.
display: flex; gap: 15px;	Activates CSS3 Flexbox layout on .cards, creating equal spacing between child card items.
flex: 1	Instructs each card to expand and distribute equally across the row.
:hover pseudo-class	Changes navigation background to orange, card background to blue, and button background to green on mouse hover.
canvas.getContext("2d")	Returns a 2D drawing context object providing methods to render shapes and text.
ctx.fillRect(50, 50, 150, 80)	Renders a solid filled blue rectangle at coordinate (50, 50) with width 150 and height 80.
ctx.arc(300, 90, 50, 0, Math.PI*2)	Constructs a full 360-degree circular arc centered at (300, 90) with radius 50, filled with red.
ctx.moveTo() & ctx.lineTo()	Defines a line segment from (430, 50) to (600, 150), drawn in green with a 5px stroke width.
ctx.fillText("B. Charishma", 250, 220)	Draws purple custom student typography on the canvas using a 30px Arial font.

8. TEST CASES AND EXECUTION DATA

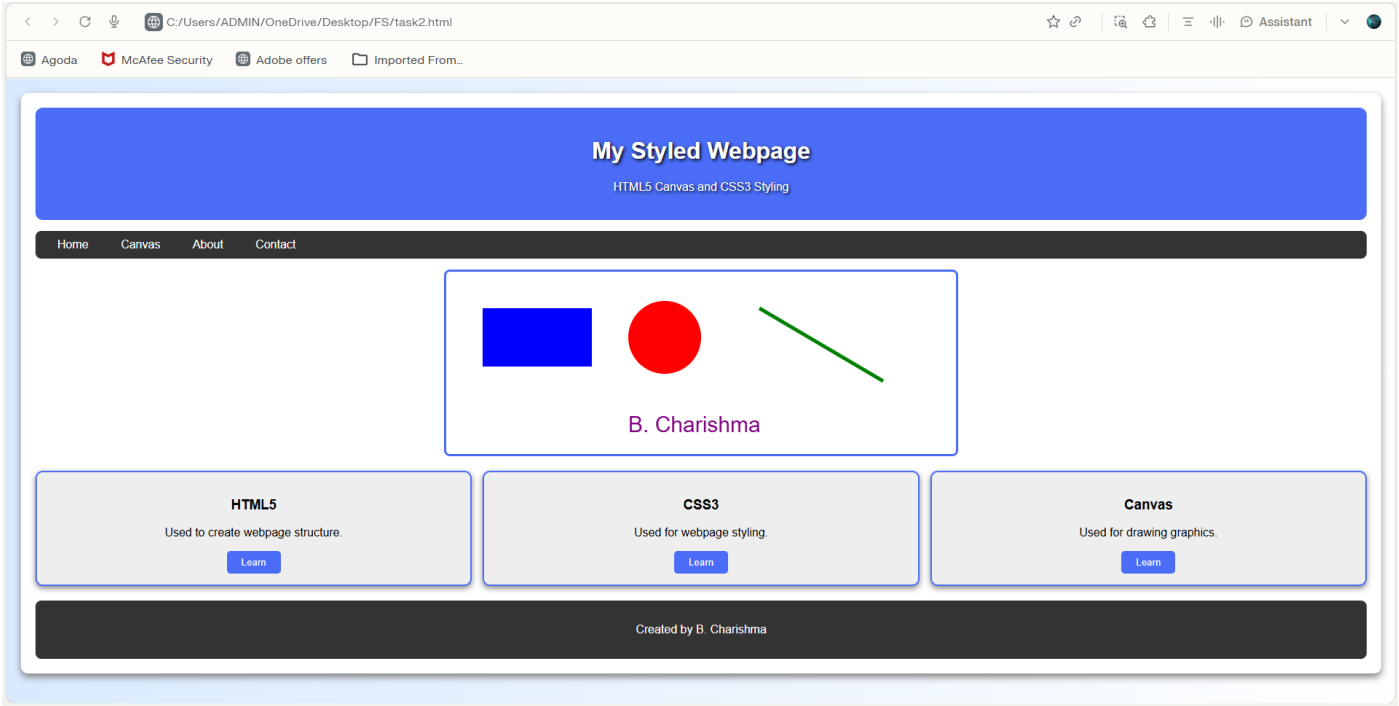
I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Module / Feature	Input / Action	Expected Result	Actual Result	Status
TC-01	Background Gradient	Load webpage	Smooth gradient from #d9eaff to white	Gradient rendered seamlessly	Pass
TC-02	Canvas Rectangle	Execute fillRect()	Blue rectangle at (50,50,150,80)	Solid blue rectangle displayed	Pass

Test Case	Module / Feature	Input / Action	Expected Result	Actual Result	Status
TC-03	Canvas Circle	Execute arc() + fill()	Red circle at (300,90) radius 50	Solid red circle displayed	Pass
TC-04	Canvas Line	Execute lineTo() + stroke()	Green diagonal line (width 5)	5px green line drawn	Pass
TC-05	Canvas Text	Execute fillText()	Purple text "B. Charishma" at (250,220)	Purple font rendered accurately	Pass
TC-06	Hover Transitions	Hover mouse on Nav, Cards, Buttons	Background shifts to orange/blue/green	Hover animations responsive	Pass

9. OUTPUT

Output: Webpage Layout with HTML5 Canvas 2D Shapes & CSS3 Cards



10. RESULT

Thus, the webpage combining **CSS3 Advanced Styling** (linear gradients, flexbox multi-column cards, text & box shadows, hover animations) and **HTML5 Canvas 2D Graphics** (rectangles, circles, lines, and custom name rendering) was successfully implemented and verified across all test cases.

